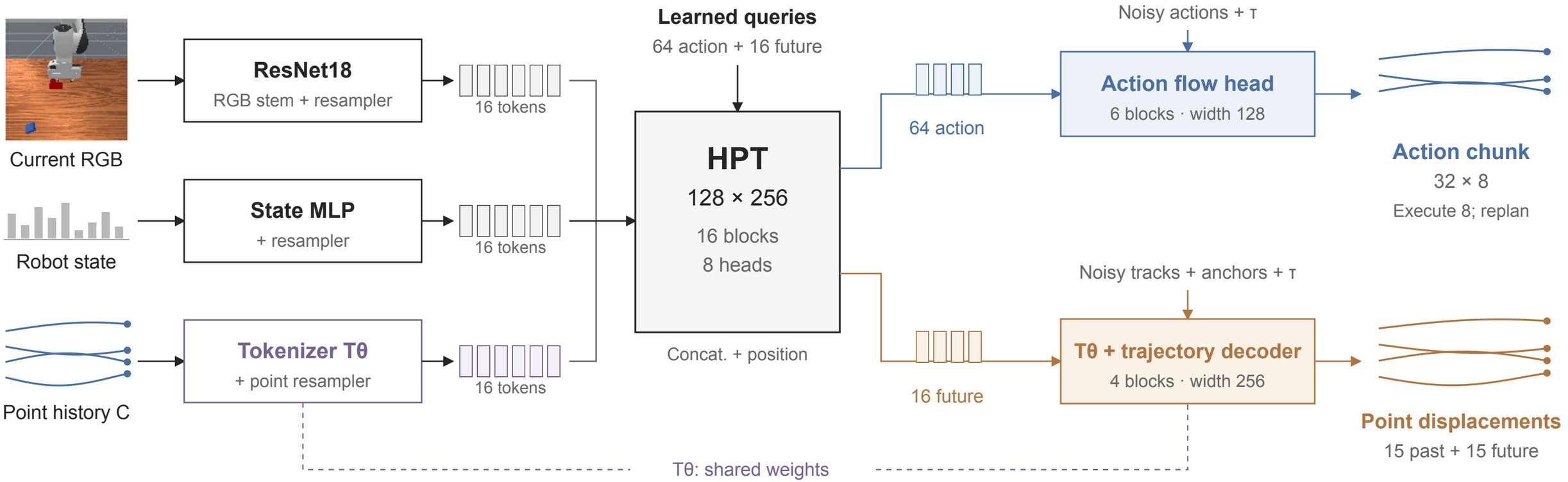
4D MAE / B5

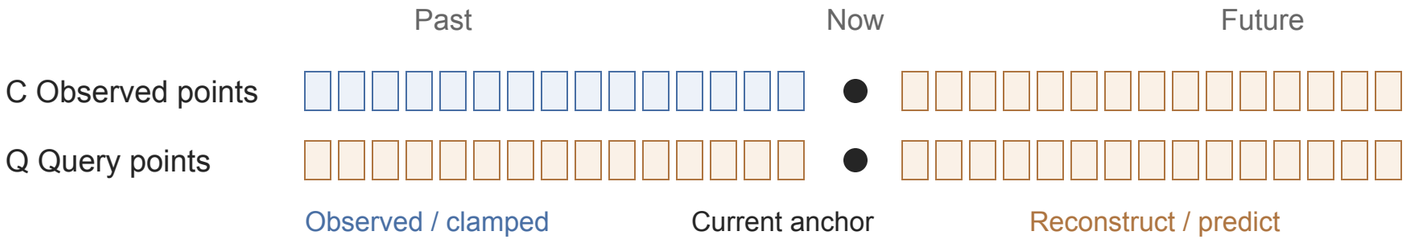
PullCube configuration

(a) Policy architecture



Control uses the action branch. Trajectory completion supplies auxiliary training supervision.

(b) Trajectory supervision



$$L = L_{action} + (L_{C,future} + L_{Q,past} + L_{Q,future}) / 3$$

Flow-matching losses on valid targets.

One decoder, called separately for C and Q.

64 C points · 64 Q points · 16 history frames · privileged simulator history in PullCube

All trajectory sketches are schematic.